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This is the accepted version of a paper published in *International Journal of Comparative Labour Law and Industrial Relations*. This paper has been peer-reviewed but does not include the final publisher proof-corrections or journal pagination.

Citation for the original published paper (version of record):

Holm, J., Quinlan, M., Williams, P., Harkema, D., Gunn, V. et al. (2026)
Algorithmic management, the AI act, and OSH risk assessments under EU law
International Journal of Comparative Labour Law and Industrial Relations, 42(3)

Access to the published version may require subscription.

N.B. When citing this work, cite the original published paper.

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Algorithmic Management, the AI Act, and OSH Risk Assessments under EU Law

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The increasingly pervasive use of digital technologies, especially algorithms and artificial intelligence (AI) in the workplace, has raised growing concerns about the need to identify and regulate their effect on occupational safety and health (OSH). The European Union (EU)'s Artificial Intelligence Act (AI Act) represents a significant regulatory framework aimed at overseeing the development and deployment of AI technologies within the EU. This paper critically examines the effectiveness of this legislation in managing occupational safety and health (OSH) risks.

While the AI Act classifies many algorithmic management (AM) systems as "high-risk" and imposes ex ante obligations on providers to assess and mitigate safety risks, its product safety-oriented approach is limited in addressing the context-specific, dynamic, and cumulative nature of workplace harms. Drawing on a doctrinal analysis of the AI Act (Regulation (EU) 2024/1689) and the EU OSH Framework Directive (89/391/EEC), this paper compares the AI Act's risk management requirements with the ongoing, participatory, and preventive duties embedded in OSH regulation. Findings highlight critical regulatory gaps, particularly limited obligations of employers ("deployers") under the AI Act and the challenges of addressing psychosocial risks through design measures alone. The article argues for a regulatory approach that aligns the AI Act with existing OSH frameworks and calls for greater guidance, oversight, and worker involvement in the deployment of algorithmic systems. Only by recognising AM as

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[**********] The authors wish to thank Aude Cefaliello, Antonio Aloisi and the reviewers for their valuable comments. We also extend our gratitude to Forte (the Swedish Research Council for Health, Working Life and Welfare) for funding the ALGOSH (Algorithmic management at work – challenges, opportunities, and strategies for occupational safety and health and wellbeing) research program (Reg. No. 2023-00015).

Holm, Johan; Quinlan, Michael; Williams, Penny; Harkema, Denise; Gunn,Virginia; Strauss, Pille; Lind, Ruben; Kalakoski, Virpi & Håkansta,Carin. 'Algorithmic Management, the AI Act, and OSH Risk Assessments under EU Law'. *International Journal of Comparative Labour Law and Industrial Relations* 42, no. 3 (2026): 1–36.
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both a technical product and a socio-organizational practice can regulation effectively safeguard worker health, safety, and dignity.

Keywords: AI Act, Algorithmic management, OSH, European Union, Risk Assessment, Product regulation, Workplace regulation, EU OSH Framework Directive (89/391/EEC), Labour law, Working environment

1 INTRODUCTION

Over the past three decades, the introduction of digital technologies has reshaped work and working conditions. One significant innovation has been algorithmic management (AM), which involves the use of artificial intelligence (AI) and other digital technologies that gather, use, or create data to enhance or automate managerial decision-making regarding work organization and worker performance.[1] While handing over of managerial functions to algorithms is done routinely within digital labour platforms, AM has expanded into traditional employment relationships in customer service, social services, manufacturing, mining, banking, transportation, logistics, and healthcare.[2] AI (which is built on algorithms) is now a central feature of many AM systems. Often used to increase efficiencies, AM affects how work activities are assigned, and how workers are hired, monitored, evaluated, rewarded, or fired.[3] AM can also affect the health and wellbeing of workers,[4] job quality and the work environment.[5]

To safeguard workers, there is substantial regulation of occupational safety and health (OSH). In the European Union (EU), the main legal instrument is the OSH

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1 P. Strauss-Raats, V. Gunn, C. Håkansta, R. Lind, J. C. Hernando Rodríguez & M. Quinlan, *Using a Modified Delphi Approach to Advance a Common Definition of Algorithmic Management* (Poster presentation at EPICOH Early Career Conference, Barcelona, Spain 2024).

2 S. Baiocco, E. Fernández-Macías, U. Rani & A. Pesole, *The Algorithmic Management of Work and Its Implications in Different Contexts* (St. Louis: Federal Reserve Bank of St. Louis 2022); I-González-Vázquez et al., Digital Monitoring, Algorithmic Management and the Platformisation of Work in Europe (Publications Office of the European Union, Luxembourg, 2025), https://data.europa.eu/doi/10.2760/9406086, JRC143072; U. Rani, A. Pesole & I. G. Vázquez, *Algorithmic Management Practices in Regular Workplaces: Case Studies in Logistics and Healthcare* (Publications Office of the European Union Luxembourg 2024), https://www.ilo.org/publications/algorithmic-management-practices-regular-workplaces-case-studies-logistics (accessed 8 Jan. 2026).

3 M. H. Jarrahi, G. Newlands, M. K. Lee, C. T. Wolf, E. Kinder & W. Sutherland, *Algorithmic Management in a Work Context*, 8(2) Big Data & Soc'y 20539517211020332 (2021), doi: 10.1177/20539517211020332; M. K. Lee, D. Kusbit, E. Metsky & L. Dabbish, *Working With Machines: The Impact of Algorithmic and Data-Driven Management on Human Workers* (Proceedings of the 33rd annual ACM conference on human factors in computing systems 2015).

4 K. H. Nilsson, T. Bodin, P. Strauss, N. Matilla-Santander, K. Badarin, E. Brulin, & C. Håkansta, *Algorithmic management is associated with psychological distress, musculoskeletal pain, and occupational accidents: a cross-sectional study in logistics,* International Archives of Occupational and Environmental Health. (2025) Dec; 98(9–10): 929–942;
H. Kinowska & L. J. Sienkiewicz, *Influence of Algorithmic Management Practices on Workplace Well-Being – Evidence from European Organisations,* 36(8) Info. Tech. & People 21–42 (2023), doi: 10.1108/ITP-02-2022-0079; E. F. Vignola, S. Baron, E. Abreu Plasencia, M. Hussein & N. Cohen, *Workers' Health under Algorithmic Management: Emerging Findings and Urgent Research Questions*, 20(2) Int'l J. Envtl. Res. & Pub. Health, 1239, doi: 10.3390/ijerph20021239.

5 Baiocco et al., *supra* n. 2; González-Vázquez et al, supra n. 2.

Framework Directive.[6] which requires the evaluation of safety and health risks to workers. However, risk assessment is not prescribed solely in OSH-regulation: the new AI Act[7] provides a mechanism to address safety issues posed by the development, implementation and use of AI systems through the requirement to establish and implement a risk management system. Consequently, when AI applications or systems are developed or used at work, there is a dual requirement to make risk assessments – one based on OSH-regulation and one based on the AI Act, which mandates risk assessments to be carried out by providers of AI systems. Given that AI is expanding the capabilities of AM systems, and AM is expanding into a wider range of workplaces, there is a need to critically examine the implications of this dual approach for mitigating AM-related OSH risks.

This study aims to examine the ways in which the AI Act regulates OSH risks arising from AM. It identifies the kind of risk assessment required under the AI Act for providers and deployers (i.e., employers) of AM systems and the role of standards and standard-setting bodies in this assessment. The paper compares how this approach may interact or overlap with established assessment frameworks for OSH-related risks under existing OSH regulations: How does the regulatory technique used in the AI Act function to mitigate OSH risks of AM? How does this differ from the OSH regulatory approach? What possible influences will the AI Act have on workplace safety risk assessments of AM? The study deals with only regulation at the EU level and focuses on regulation applicable to standard employment, excluding platform work regulation.[8]

Legal doctrinal method is used to study the legal responsibility to perform risk assessments of providers of AI systems and users of AM systems through an analysis of the sources of law. These sources include EU legislation, preparatory works, and case law. The main sources studied are the AI Act and workplace safety regulations, such as the Framework Directive. Other types of regulation that may address OSH risks are excluded (e.g., privacy regulation, working time regulation, and collective worker rights).

The paper is divided into nine sections including this introductory section. In the next section (2), we examine what AM is and what is known about its OSH effects. Section 3 further sets the context by contrasting product safety regulation and workplace (OSH) regulation. Section 4 then examines the role of risk assessment across these regulatory regimes. Sections 5 and 6 introduce the AI Act, explain its relevance to employment, and set out the Act's risk management and

[6] Council Directive of 12 Jun. 1989 on the introduction of measures to encourage improvements in the safety and health of workers at work (89/391/EEC).
[7] Regulation (EU) 2024/1689.
[8] Such as the Platform Work Directive (EU) 2024/2831 which regulates some aspects of AM in Ch. III.

risk assessment requirements. Section 7 identifies the limitations of a product safety–focused approach to OSH. Section 8 develops the labour law perspective, explaining how OSH frameworks can complement the AI Act's approach. A concluding section (9) then draws the key findings together.

2 WHAT IS AM AND WHAT ARE THE AM-RELATED OSH HAZARDS TO WORKERS?

Exemplified by digital platforms in the gig economy, Algorithmic management was first defined as the assumption of managerial tasks by digital programmes.[9] Acknowledging that AM also occurs in traditional employment settings, later definitions include the use of a diverse range of tools, (including AI) to automate the coordination, control and evaluation of workers.[10] Across wide-ranging work contexts, algorithms automate and augment management functions commonly carried out by humans, such as: planning; scheduling; goalsetting; monitoring location, productivity, and output; evaluating performance; instructing, rewarding, and sanctioning workers; and in hiring and firing processes.[11] While a human supervisor's ability to monitor and assess worker performance is limited, digital technologies can track and direct workers individually in real time on a larger scale, particularly when AI is incorporated. By automating delegated managerial authority AM embeds and extends the scope and scale of management prerogatives while reducing human intervention and oversight.[12]

AM systems augmented with AI may have paradoxical consequences.[13] These include the potential to both reduce and reproduce bias and discriminatory practices, to simultaneously enable and control workers, and to alleviate or

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[9] Lee et al., *supra* n. 3.

[10] Baiocco et al., *supra* n. 2; E. Fernández-Macías, C. Urzì Brancati, S. Wright & A. Pesole, *The Platformization of Work, Evidence from the JRC Algorithmic Management and Platform Work Survey* (AMPWork) (Publications Office of the European Union 2023), https://publications.jrc.ec.europa.eu/repository/handle/JRC133016 (accessed 8 Jan.2026); A. Mateescu & A. Nguyen, *Algorithmic Management in the Workplace*, Data & Soc'y (2019).https://datasociety.net/wp-content/uploads/2019/02/DS_Algorithmic_Management_Explainer.pdf (accessed 8 Jan. 2026).; K. H. Nilsson, N. Matilla-Santander, M. K. Lee, E. Brulin, T. Bodin & C. Håkansta, *Algorithmic Management and Occupational Health: A Comparative Case Study of Organizational Practices in Logistics*, Safety Sci. (2025) 106863, doi: 10.1016/j.ssci.2025.106863.

[11] Baiocco et al., *supra* n. 2; K. C. Kellogg, M. A. Valentine & A. Christin, *Algorithms at Work: The New Contested Terrain of Control*, 14(1) Acad. Mgmt. Annals 366–410 (2020), doi: 10.5465/annals.2018.0174.

[12] A. Aloisi & V. De Stefano, *Essential Jobs, Remote Work and Digital Surveillance: Addressing the COVID-19 Pandemic Panopticon*, 161(2) Int'l Lab. Rev. 289–314 (2022), doi: 10.1111/ilr.12219.

[13] A. Cefaliello, P. V. Moore & R. Donoghue, *Making Algorithmic Management Safe and Healthy for Workers: Addressing Psychosocial Risks in New Legal Provisions*, 14(2) Eur. Lab. L.J. 192–210 (2023), doi: 10.1177/203195252311674; J. Holmström & M. Hällgren, *AI Management Beyond the Hype: Exploring the Co-constitution of AI and Organizational Context*, 37(4) AI & Soc'y 1575–1585 (2022), doi: 10.1007/s00146-021-01249-2.

exacerbate workload, stress and strain.[14] The impact of AM on working conditions and OSH largely depends on preexisting organizational and regulatory contexts and interventions that aim to eliminate the negative effects and strengthen AM's positive potential. For example, Jensen et al. and Nilsson et al., found that worker involvement and transparency in organizational decision-making could reduce negative implications of AM on workers and job quality.[15] Bender and Söderqvist argued that the use of AM can increase safety in mining without invading worker privacy and autonomy when workers and managers share common goals.[16] Nilsson et al. further show that negative AM effects could be tempered by systematic OSH management.[17] These findings suggest that, with proper risk assessment, hazard prevention, and worker co-determination, OSH benefits can arise from AM.

Considerable research and policy debate on the OSH effects of AM have focused on digital platform work, where there is ample evidence of health-related impacts of AM, including safety and occupational wellbeing.[18] For example, in the driving and food delivery sectors, digital platforms such as Uber, use AM to distribute work, control pricing, and monitor worker ratings, prompting workers to work long hours or continue working when sick or injured.[19] Work intensification exacerbated by job insecurity and low/irregular pay has been found to undermine the OSH of precarious workers more generally.[20] Todolí-Signes categorises the occupational risks associated with AM as: 1) constant monitoring;

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14 Cefaliello et al., *supra* n. 13; Nilsson et al., *supra* n. 10; N. Noponen, P. Feshchenko, T. Auvinen, V. Luoma-aho & P. Abrahamsson, *Taylorism on Steroids or Enabling Autonomy? A Systematic Review of Algorithmic Management*, 74(3) Mgmt. Rev. Q. 1695–1721 (2024), doi: 10.1007/s11301-023-00345-5.

15 M. Thorn Jensen, G. Rinse Oosterwijk & A. Sonne Nørgaard, *Computer in Command: consequencses of Algorithmic Management for Workers* (2024), feps-europe.eu/wp-content/uploads/2024/06/Policy-Study-Int-Report-Algorithmic-Management-web-PP.pdf, The Oxford Handbook of Regulation (2010; online ed., Oxford Academic, 2 Sep. 2010),https://doi.org/10.1093/oxfordhb/9780199560219.003.0014 (accessed 8 Jan. 2026); Nilsson et al., *supra* n. 10.

16 G. Bender & C. F. Söderqvist, Human-Centered or Biorobotized Automation? Technological Codetermination in an Innovative Mining Company, in Essays on Exit, Voice, and Technology. Industrial Relations in Modern Swedish Labour Markets (C. F. Söderqvist ed., Department of Industrial Economics, Blekinge Institute of Technology 2024), https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1858226&dswid=9841 (acccessed 8 Jan. 2026).

17 Nilsson et al., supra n. 10.

18 Vignola et al. *supra* n. 4; Kinowska & Sienkiewicz, *supra* n. 4; E. Perez Vallejos, L. Dowthwaite, H. Creswich, V. Portillo, A. Koene, M. Jirotka, A. McCarthy & D. McAuley, *The Impact of Algorithmic Decision-Making Processes on Young People's Well-Being*, 27(1) Health Informatics J. 1460458220972750 (2021), doi: 1460458220972750.

19 E. L. Bucher, P. K. Schou & M. Waldkirch, *Pacifying the Algorithm – Anticipatory Compliance in the Face of Algorithmic Management in the Gig Economy*, 28(1) Org. 44–67 (2021), doi: 10.1177/1350508420961531; I. Kadolkar, S. Kepes & M. Subramony, *Algorithmic Management in the Gig Economy: A Systematic Review and Research Integration*, J. Organizational Behav. 1057–1080 (2024), doi: 10.1002/job.2831.

20 E. Bluff, R. Johnstone & M. Quinlan, *Regulating Health and Safety in Work for Digital Labor Platforms in Australia: The Example of Food Deliverers* 1(1) J. Work Health & Safety Reg. 92–116 (2023), doi: 10.57523/jaohlev.21–007.

2) work intensification; 3) lack of autonomy; 4) algorithmic bias and discrimination; 5) complexity and lack of transparency; and 6) malfunctions and ethical concerns.[21] These aspects represent interrelated features of AM leading to physical and psychosocial effects on workers. For example, while AM may support autonomous work, the associated surveillance and constant monitoring can negatively impact the physical and psychosocial health of workers.[22]

Constant monitoring can contribute to work intensification leading to injuries and physical harms such as over-exertion and other forms of ill health.[23] In warehousing and logistics, workplace surveillance has been linked to increased physical exertion.[24] In the widely cited U.S. case of Amazon warehousing, AM enacted through productivity monitoring devices, caused workers to work at unsustainable pace and take fewer breaks.[25] In workplaces with robots or large machinery, malfunctions of AM may result in poor ergonomics and place workers in direct physical harm, with already documented fatalities.[26]

AM-related changes to the psychosocial work environment can also have harmful effects by contributing to worker stress and burnout. For example, AM technologies for monitoring, speech analytics, scheduling, and automated real-time coaching in contact centres have been linked to higher levels of stress, customer mistreatment, and job insecurity.[27] Similar experiences have been reported for

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21 A. Todolí-Signes, *Making Algorithms Safe for Workers: Occupational Risks Associated With Work Managed by Artificial Intelligence*, 27(4) Transfer Eur. Rev. Lab. & Res. 433–452 (2021), doi: 10.1177/10242589211035040.

22 M. Flyverbom, *Transparency: Mediation and the Management of Visibilities*, 10(1) Int'l J. Comm. 110–122 (2016), https://doi.org/http://ijoc.org/index.php/ijoc/article/download/4490/1531; Noponen et al., *supra* n. 14; P. Williams & M. H. Khan, *Framing Algorithmic Management: Constructed Antagonism on HR Technology Websites*, 40(1) New Tech. Work & Emp. 102–123 (2024), doi: 10.1111/ntwe.12305.

23 *See* J. Berg, *Protecting Workers in the Digital Age: Technology, Outsourcing, and the Growing Precariousness of Work*, Comp. Lab. L. & Pol'y J. 69, 78 (2019), doi: 10.2139/ssrn.3413740. for summary.

24 A. Delfanti, *Machinic Dispossession and Augmented Despotism: Digital Work in an Amazon Warehouse*, 23(1) New Media & Soc'y 39–55 (2021), doi: 10.1177/1461444819891613; F. Lorson, A. Fügener & A. Hübner, *New Team Mates in the Warehouse: Human Interactions With Automated and Robotized Systems*, 55(5) Iise Transactions 536–553 (2023), doi: 10.1080/24725854.2022.2072545; A. Mateescu & A. Nguyen, *Algorithmic Management in the Workplace*, Data & Soc'y (2019). https://datasociety.net/wp-content/uploads/2019/02/DS_Algorithmic_Management_Explainer.pdf (accessed 8 Jan. 2026).; Nilsson et al., *supra* n. 10.

25 Delfanti, *supra* n. 24; E. Saner, *Employers are Monitoring Computers, Toilet Breaks – Even Emotions. Is Your Boss Watching You* (The Guardian 14 May 2018).; A. J. Wood, *Algorithmic Management: Consequences for Work Organisation and Working Conditions* (St. Louis: Federal Reserve Bank of St. Louis 2021).

26 H. Jackson, *Recurrent Causes of Occupational Incidents in the NSW, Australia Coal and Metalliferous Mining Industry* (Doctor of Philosophy, University of Newcastle, NOVA 2021): R. Burgess-Limerick, T. Horberry, D. Lynas, A. Hill & J. M. Haight, *Human Aspects of Automation and New Technology in Mining: Integrating People and Technology Through Human-Centred Design*, vol. 1, 1–8 CRC Press (1st ed., A White Paper prepared for ACARP project C#4026, University of Queensland, St. Lucia 2024), https://doi.org/10.1201/9781003380887-1.

27 V. Doellgast, S. O'Brady, J. Kim, D. Walters, A. Acevedo, N. Dragsbaek, T. Hegeman, S. Ivanovski, N. McCormick, A. Rana, M. Snyder, J. Starcevic, I. Wagner & D. Kylasam Iyer, *AI in Contact Centers: Artificial Intelligence and Algorithmic Management in Frontline Service Workplaces* (2023), https://doi.org/10.13140/RG.2.2.36641.58724; L. Layne, Robot-related fatalities in the US 1992-2017, American Journal of Industrial Medicine (2023) 66(6)454-61.

workers in desk-based occupations,[28] and manufacturing.[29] AM may contribute to diminished job quality and worker motivation by constraining worker autonomy, increasing task complexity, and transferring supervisory and social support to AM systems.[30] A Nordic country study found negative associations between AM use and OSH and well-being indicators, including increased workload and pace, job insecurity, autonomy, satisfaction, motivation, stress levels, and trust.[31] At the same time, when used appropriately, AM can facilitate OSH improvements. This may occur through improved tracking in high-risk situations, detection and automated correction of unsafe practices, remote control of equipment to remove workers from hazardous environments, and the use of AI for early warning systems or to better identify precursors to hazardous events.[32] Risk assessments must therefore enable organizations to capitalize on the benefits that AI and AM may offer.

The hazards associated with AM and AI are neither new nor surprising in most instances. However, they present additional challenges for unions and regulators because with AM underlying parameters of the changes to work organization and processes become opaque. AM fundamentally restructures work organization, control, and power relations at work and is best understood not as a tool with inherently hazardous or beneficial qualities, but as a sociotechnical system with social, regulatory, and technological dimensions. AM is not used in the same way in platform and traditional employment settings,[33] in the global north and the global south,[34] in white-collar and blue-collar industries,[35] or even

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[28] Baiocco et al., *supra* n. 2.

[29] M. Dupuis, *Algorithmic Management and Control at Work in a Manufacturing Sector: Workplace Regime, Union Power and Shopfloor Conflict over Digitalisation*, 40(1) New Tech. Work & Emp't 81–101 (2025), doi: 10.1111/ntwe.12298.

[30] Todolí-Signes, *supra* n. 21; Vignola et al., *supra* n. 4; P. Williams, P. McDonald & R. Mayes, *The Growing 'Gig Economy': Implications for the Health and Safety of Digital Platform Workers*, in *Handbook on Management and Employment Practices. Handbook Series in Occupational Health Sciences*, vol 3 769–785 (P. Brough, E. Gardiner & K. Daniels eds, Springer 2022), https://doi.org/10.1007/978-3-030-29010-8_32; Wood, *supra* n. 25.

[31] Thorn Jensen et al., *supra* n. 15.

[32] H. Luo & S. G. Paal, *Artificial Intelligence-Enhanced Seismic Response Prediction of Reinforced Concrete Frames*, Advanced Eng'g Informatics 101568 (2022), doi: 10.1016/j.aei.2022.101568; C. A. Perez, C. M. Aravena, D. Schulz, L. Cament, L. Castillo, F. Smith & P. A. Estevez, *Automatic Safety Video Surveillance-Tracking System to Avoid Vehicle-Workers Interaction for Mining Applications* (International Symposium on Optomechatronic Technologies (ISOT) 2014).

[33] I. Lippert, K. Kirchner & M. Wiener, Context Matters: The Use of Algorithmic Management Mechanisms in Platform, Hybrid, and Traditional Work Contexts, Proceedings of the 56th Hawaii International Conference on System Sciences 5282 (2023).

[34] U. Rani, A. Pesole & I. G. , Algorithmic Management Practices in Regular Workplaces: Case Studies in Logistics and Healthcare (Publications Office of the European Union Luxembourg 2024). https://www.ilo.org/publications/algorithmic-management-practices-regular-workplaces-case-studies-logistics (accessed 8 Jan. 2026).

[35] I-González-Vázquez et al., Digital Monitoring, Algorithmic Management and the Platformisation of Work in Europe (Publications Office of the European Union, Luxembourg, 2025), https://data.europa.eu/doi/10.2760/9406086, JRC143072.; A. Milanez, A. Lemmens and C. Ruggiu "Algorithmic

between companies within the same regulatory environment[36] and the outcomes vary widely. These differences demonstrate that the OSH risks arising from a particular AM system are contingent on the pre-existing work environment, and the approach to embedding AM in the organization. This makes it difficult for prevailing assessment frameworks to identify AM-related OSH impacts. An OSH hazard refers to something – be that physical, psychosocial or some combination of both – that can harm the safety, health or wellbeing of workers. In AM contexts, hazards are the inherent system features, such as continuous individualized surveillance, opaque decision-making logic, constantly changing performance targets, and automated behavioural nudging, that have the potential to undermine workers' physical or psychosocial wellbeing. These hazardous characteristics exist even when no adverse event has yet occurred and often arise from AM's ordinary functioning, not malfunction. Hazards associated with digital work systems have been the subject of attention for some time.[37] The pace of change and rapid adoption of AM and AI across sectors however makes it difficult for regulators and policy makers to obtain information on the number of affected workers and the nature and extent of the associated hazards.[38] By contrast, risks involve attempts to estimate the likelihood that these hazards will result in negative worker outcomes within a given scenario. Risk assessments are central to the regulatory regimes of both product safety and workplace safety. Risk is the possibility of a future event and its consequences in relation to something valued.[39] Risk is commonly framed in negative terms, emphasizing a potential loss or a failure to achieve desired objectives.[40] Failure to act can also constitute a risk, representing a missed

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management in the workplace: New evidence from an OECD employer survey", OECD Artificial Intelligence Papers, No. 31, (OECD Publishing, Paris, 2025), https://doi.org/10.1787/287c13c4-en.

36 K. H. Nilsson, N. Matilla-Santander, M. K. Lee, E. Brulin, T. Bodin & C. Håkansta, Algorithmic Management and Occupational Health: A Comparative Case Study of Organizational Practices in Logistics, Safety Sci. 187 (2025) 106863, doi: 10.1016/j.ssci.2025.106863

37 P. Moore The Threat of Physical and Psychosocial Violence and Harassment in Digitalized Work (International Labour Office, Geneva 2018); European Agency for Safety and Health at Work: OSH and the future of work: Benefits and risks of artificial intelligence tools in workplaces, Discussion Paper, Bilbao (2019).

38 A. Paliokaitė, A. Christenko & A. Aloisi et al., Study Exploring the Context, Challenges, Opportunities, and Trends in Algorithmic Management in the Workplace – Final Report (European Commission: Directorate-General for Employment, Social Affairs and Inclusion, Visionary Analytics, Publications Office of the European Union 2025), https://data.europa.eu/doi/10.2767/5629841 (accessed 8 Jan. 2026).

39 T. Aven, Risk Assessment and Risk Management: Review of Recent Advances on Their Foundation, 253(1) Eur. J. Operational Res. 1–13 (2016), doi: 10.1016/j.ejor.2015.12.023; J. Black, The Role of Risk in Regulatory Processes, in R. Baldwin, M. Cave, and M. Lodge (eds). The Oxford Handbook of Regulation (2010; online edn, Oxford Academic, 2 Sept. 2010), https://doi.org/10.1093/oxfordhb/9780199560219.003.0014 (accessed 8 Jan. 2026.)

40 J. Black, The Role of Risk in Regulatory Processes, in R. Baldwin, M. Cave, and M. Lodge (eds). The Oxford Handbook of Regulation (2010; online edn, Oxford Academic, 2 Sept. 2010), https://doi.org/10.1093/oxfordhb/9780199560219.003.0014 (accessed 8 Jan. 2026.)

opportunity for potential gains.[41] Quantitatively, risk can be defined as a combination of the probability and severity of adverse outcomes.[42] While traditional risk assessment tools are a good fit to review discrete, measurable events (e.g., acute injury due to work pace changes), they struggle to quantify emergent exposures such as chronic stress, erosion of autonomy, or the incremental intensification of work. Harms, in contrast, refer to the actual negative outcomes experienced by workers, individually or collectively, and these often emerge gradually within AM-mediated work environments. Long-term harms – such as cumulative fatigue, sustained job insecurity, or systemic bias exacerbated through automated evaluation – are typically the result of sustained exposure to AM hazards interacting with organizational pressures. Because conformity assessments evaluate compliance at a point in time, and risk assessments rely on probabilistic modelling of foreseeable events, both frameworks tend to overlook these structural, and cumulative harms. For policymakers and OSH practitioners, this suggests the need for hazard sensitive and context aware assessment approaches that foreground the sociotechnical embedding of AM, incorporate worker participation, and enable longitudinal monitoring of exposures. Such approaches reflect the ways in which AM reshapes work organization and power relations and can help ensure that gradual or systemic harms do not fall outside regulatory visibility.

3 PRODUCT SAFETY REGULATION AND WORKPLACE REGULATION

Worker safety is achieved both by regulation aimed at ensuring that the tools and machines used at work are safe to use (product safety regulation) and regulation aimed at guaranteeing that the work is organized and carried out safely (workplace regulation).

The objective of product safety regulation is not only to protect end-consumers, but also workers operating or exposed to these products, by ensuring hazards are engineered out *ex ante* before the product is used in any workplace or environment.[43] Responsibility is placed on manufacturers to eliminate or minimize hazards in the product's design and production stages. These regulations aim to prevent injuries by requiring equipment, machinery, or consumer product to meet essential safety requirements. For example, the EU Machinery Directive (2006/42/EC) sets

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41 F. A. Sortino & S. Satchell (Eds), Managing Downside Risk in Financial Markets: Theory, Practice and Implementation (Butterworth-Heinemann 2001).

42 T. Aven, Risk Assessment and Risk Management: Review of Recent Advances on Their Foundation, 253(1) Eur. J. Operational Res. 1–13 (2016), doi: 10.1016/j.ejor.2015.12.023.

43 R. Baldwin, M. Cave & M. Lodge, *Understanding Regulation: Theory, Strategy, and Practice* (Oxford University Press 2011).

standardized health and safety requirements for machinery to protect people interacting with machines.[44] This directive aims to protect EU workers and citizens through harmonized product design standards. In practice, compliance often involves meeting technical safety standards, e.g., guarding dangerous parts, fail-safes, warning labels, and affixing a conformity mark (such as CE marking in EU) to indicate a product's adherence to safety requirements.[45]

In contrast, OSH regulation is predominantly process-based, focusing on workers within the specific context of their working environment. These laws impose on employers (and other parties) the duty to maintain safe and healthy working conditions, addressing risks during the use of equipment or performance of work.[46] The hazards must be managed in the workplace through preventive measures, training, and safety policies tailored to the situation.[47] An example is the OSH Framework Directive, which lays down minimum requirements and general basic principles for the prevention of work-related hazards. In essence, OSH regulation creates a general duty for employers to ensure the safety and health of their workers in every aspect related to work, including safe work equipment, safe processes, training and supervision, and emergency preparedness. The objective is to continuously identify and mitigate risks arising from the specific conditions of work, thereby protecting workers on an ongoing basis.

Both regulatory approaches share the goal of preventing harm but operate at different stages and through different actors – one through safe product design, the other through safe work practices. This presents a problem of external coherence in EU law. Where distinct regulatory regimes pursue related protective objectives – here, prevention of harm to human health and safety – they should be designed and applied so that one regime does not, in practice, undermine the other. In the OSH Framework Directive, the employer bears a broad duty to ensure workers' safety and health "in every aspect related to work",[48] operationalized through prevention principles, risk assessment, and worker consultation and participation. By contrast, EU product safety regulation primarily seeks to prevent unsafe products reaching the market through ex ante design stage risk management and conformity mechanisms. The AI Act largely follows this product logic for high-risk AI systems through provider centred requirements and conformity assessment.

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44 The Machinery directive is repealed by the Regulation (EU) 2023/1230 on machinery. The Regulation applies from 20 Jan. 2027.

45 S. Weatherill, *EU Consumer Law and Policy* (Edward Elgar Publishing 2014).

46 P. James & D. Walters, *Regulating Health and Safety at Work: An Agenda for Change?* (Institute of Employment Rights 2005).

47 D. Walters, R. Johnstone, K. Frick, M. Quinlan, G. Baril-Gingras & A. Thebaud-Mony. *Regulating Workplace Risks A Comparative Study of Inspection Regimes in Times of Change* (Edward Elgar Publishing 2011).

48 Article 5(1) Directive 89/391/EEC.

4 RISK ASSESSMENTS OF PRODUCTS AND WORKPLACE RISK

Product-safety risks concern the potential for a specific system or device to cause harm when it malfunctions or is used improperly, focusing on defects, failures, or unsafe design features. By contrast, AI as a potentially risky technology, encompasses the broader socio-technical capacities of AI systems, such as opacity, adaptivity, error proliferation, and data-driven bias, that may generate unpredictable or systemic impacts beyond what can be attributed to any single product or system component. Workplace OSH risks differ yet again: they arise through the interaction between these technological capacities and the organization of work, shaping day-to-day exposures that influence workers' physical, psychosocial, and organizational wellbeing. In other words, AM-enabled AI technologies may not present immediate product-safety hazards in themselves but can still transform work processes in ways that intensify pace, reduce autonomy, or institutionalize surveillance, thereby generating cumulative workplace OSH risks.

In practice, risk assessment does not simply entail the measurement of risk (including the effectiveness of existing control measures) but managing risk by implementing enhanced controls, monitoring their effectiveness and modifying controls as required. Managing risks is connected to uncertainties and possibilities of controlling the future.[49] The limitations arising from the *unknown unknowns* are particularly relevant to modern AI applications, where the behaviour of systems may be unpredictable.[50]

Managing uncertainty by controlling risks, both low and high, in a society is often regarded as an inherent goal of regulation.[51] In recent decades, reliance on risk control and risk management as tools has gradually increased among private organizations operating in increasingly fragmented regulatory regimes, where regulatory activities must be internalised and self-imposed, despite a range of barriers to adherence.[52] However, reliance on regulation through risk assessment in areas like environmental protection, engineering, financial services or public health, demands both robust risk management methods and risk literacy among various actors.

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[49] A. Giddens, *Risk and Responsibility*, 62(1) Mod. L. Rev. 1 (1999), doi: 10.1111/1468-2230.00188; O. E. Igbekoyi, O. L. Oluwagbade & M. E. Dagunduro, *Risk Management and Uncertainty Interplay*, 9(6) Int'l J. Econ. & Fin. Mgmt. 188–198 (2024), doi: 10.56201/ijefm.v9.no6.2024.pg188.198.

[50] Sortino & Satchell, *supra* n. 41; R. Paul, *European Artificial Intelligence 'Trusted Throughout the World': Risk-Based Regulation and the Fashioning of a Competitive Common AI Market*, 18(4) Reg. & Governance 1065–082 (2024), doi: 10.1111/rego.12563.

[51] J. Black, *The Role of Risk in Regulatory Processes*, in R Baldwin, M. Cave, and M. Lodge (eds). *The Oxford Handbook of Regulation* (2010; online edn, Oxford Academic, 2 Sept. 2010), https://doi.org/10.1093/oxfordhb/9780199560219.003.0014 (accessed 8 Jan. 2026.); J. Black & R. Baldwin, *When Risk-Based Regulation Aims Low: Approaches and Challenges*, 6(1) Reg. & Governance 2–22 (2012), doi: 10.1111/j.1748–5991.2011.01124.x.

[52] R. Baldwin, M. Cave & M. Lodge, *Understanding Regulation: Theory, Strategy, and Practice* (Oxford University Press 2011).; B. Refslund, *How Self-Regulating Can It Be? Explaining Limitations in Firms'compliance With Reflexive Regulation,* 4(4:1) Int'l Rev. Pub. Pol'y 55–113 (2022), doi: 10.4000/irpp.2590.

4.1 Risk assessment of product safety

In the realm of product safety, risk assessment is predominantly an *ex ante* process, undertaken by manufacturers prior to market introduction. For instance, the European Union's Machinery Directive (2006/42/EC) mandates that manufacturers conduct a comprehensive risk assessment for each machine, identifying potential hazards throughout its lifecycle including construction, usage, and maintenance, estimating the severity and probability of these risks, and integrating appropriate risk-reduction measures into the design.[53] A technical file and conformity assessment report documents the machine's compliance with essential safety requirements. This process ensures the product's design incorporates safety measures such as eliminating sharp edges or using safer materials, as well as safeguards like emergency stop devices and warning systems. Only after mitigating hazards to an acceptable level, often verified through testing or third-party certification, can the product be introduced to the market.[54]

Overarching product safety legislation, such as the EU's General Product Safety Regulation (2023/988), extends risk assessment beyond the design phase to encompass the entire product lifecycle. This requires manufacturers and other economic operators to continuously monitor and address emerging risks, utilizing methods like safety data analysis and initiating recalls if new hazards are identified.[55] Thus, within the product safety framework, risk assessment functions as a proactive, conformity assessment exercise aimed at preventing unsafe products from reaching the market.

4.2 Risk assessment of workplace safety

In the context of workplace safety, risk assessment is a continuous, systemic and proactive process central to an employer's obligation to ensure a safe working environment. These assessments guide employers in determining necessary safety measures tailored to their specific workplace context. Unlike the singular, pre-market focus in product safety, workplace risk assessments are continuous and situational, requiring employers to regularly evaluate their specific work environments to identify and mitigate hazards to which employees may be exposed, even if the implementation of workplace risk assessment can vary between different jurisdictions.[56] Employers' assessments must iteratively account for contextual

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53 S. Weatherill, *EU Consumer Law and Policy* (Edward Elgar Publishing 2014).

54 T. Klindt, T. Kraus, D. von Locquenghien & H. J. Ostermann, *The New EC Machinery Directive 2006* (Beuth Verlag GmbH 2007), https://books.google.ro/books?id=j4A4Ef4LO2cC (accessed 8 Jan. 2026).

55 Weatherill, *supra* n. 53.

56 A. C. J. M. Wilthagen, A. Karageorgiou, P. L. Jensen & D. Walters, *Risk Assessment in Four Member States of the European Union*, in *Systematic Occupational Health and Safety Management – Perspectives on an*

factors such as the experience levels of workers, facility layout, concurrent operation of multiple machines, maintenance schedules, and environmental conditions. For example, the Framework Directive obligates employers to perform comprehensive risk assessments encompassing all workplace hazards, including those related to psychosocial factors, and to keep these assessments up-to-date, especially considering workplace modifications.[57] The involvement of workers and/or worker representatives in "all questions relating to safety and health at work" is required.[58] It complements product safety assessments by addressing risks that could not be eliminated during the design phase or new risks introduced by specific work situations.

In many jurisdictions, legislation explicitly integrates these two assessment processes: employers are expected to utilize work equipment that is safe by design and conduct their own risk evaluations to ensure proper integration and safe use within their specific work environment.[59] Practically, this means that even if a machine bears a "CE" mark (indicating that it is approved for the EU-market for a specified purpose) or other safety certification, employers must assess its application in their workplace and, if necessary, implement additional measures, such as extra guarding or safety signage. Thus, both regulatory frameworks employ risk assessments, *ex ante* conformity assessments in product safety, and site-specific evaluations in workplace safety, as tools to safeguard individuals from harm. Distinctions emerge in the timing, scope, and the responsible party: manufacturers focus on general product hazards prior to use, while employers address all hazards within their specific workplace on an ongoing basis.

Both regulatory regimes are relevant for the regulation of AM. AM that is carried out by AI systems is regulated as a product under the AI act and at the same time as a practice under workplace safety regulation.

5 INTRODUCTION TO THE AI ACT

The European Union's AI Act[60] is a regulatory framework aimed at overseeing the development and deployment of AI technologies within the EU. The AI Act was

International Development 251–284 (K. Frick, P. L. Jensen, M. Quinlan & T. Wilthagen eds, Pergamon, Elsevier Science 2000).

57 Article 6 Directive 89/391/EEC.

58 Article 11(1) Framework Directive.

59 R. Nielsen, *EU-arbejdsret.(6.udgave)* (København: Jurist- og Økonomforbundet Forlag 2024).

60 Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 Jun. 2024 laying down harmonized rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act).

adopted on June 13 2024, and enters into force on August 2 2026.[61] Its legal basis is derived from the right of the EU to adopt measures to ensure the establishment and functioning of the internal market.[62] The purpose of the regulation is twofold. First, it aims to improve the functioning of the internal market by harmonizing the rules for developing, placing into the market, and using AI systems. Secondly, it aims to protect against the harmful effects of AI systems and ensure a high level of protection of health, safety, and fundamental rights.[63]

The AI Act regulates AI systems defined as:

> a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.[64]

Although many AM systems will fall under the definition of the AI Act, others are excluded, such as "systems that are based on the rules defined solely by natural persons to automatically execute operations".[65] This means that simple, preprogrammed AM systems will fall outside the scope of the AI Act.

The AI Act adopts a risk-based and cross-sectoral approach to AI governance.[66] Rather than focusing on specific areas such as healthcare or finance in separate laws, the AI Act provides a horizontal framework that applies to AI systems in general, with obligations calibrated to the level of risk posed by a given system's intended use. At the highest level, the Act defines a category of "unacceptable risk" as follows: AI uses that are prohibited outright, for instance, social scoring by governments or certain forms of manipulative AI that can cause harm.[67] Next, it designates certain AI systems as "high-risk" and subjects them to comprehensive requirements before they can be placed on the market. High-risk systems are those posing "significant risks to the health and safety or fundamental rights of persons" by their intended purpose and context of use.[68] AI systems related to safety components of regulated products such as AI applications integrated into medical devices, lifts or vehicles are

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61 Article 113 AI Act. However, some parts of the act enter into force at other dates, *see* Art. 113 (a)–(c). When different parts of the Act enters into force will probably be affected by the Digital Omnibus Regulation Proposal, COM (2025) 837.

62 Article 114 Treaty on the Functioning of the European Union (TFEU).

63 Recital 1 AI Act.

64 Article 3 (1) AI Act.

65 Recital 12 AI Act.

66 For a general overview of the risk based appoach to AI regulation, *see* M. E. Kaminski, *Regulating the Risks of AI*, Boston University Law Review 103 (2023) 1347-1441.

67 Article 5 AI Act.

68 European Commission, Explanatory Memorandum to COM (2021) 206 – Harmonised Final, Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts, COM (2021) 206 final (22 Apr. 2021), para. 1.1.

classified as high-risk.[69] Additionally, AI applications in sensitive domains, such as critical infrastructure, education, law enforcement, and, (of particular interest here), employment, are classified as high-risk.[70] AI systems not falling into the high-risk category are either categorized as "limited risk" with minimal transparency obligations or "minimal risk" which are largely unregulated in the AI Act. In this way, the AI Act functions like a product safety regime: higher-risk products must meet more rigorous standards (and undergo compliance checks) before they can be offered in the EU market.[71]

The AI Act places significant obligations on providers (developers) of high-risk AI systems. Providers must, inter alia, implement a risk management system, maintain technical documentation, conduct conformity assessments, and register high-risk AI systems in an EU database.[72] That most requirements are placed on the providers is a function of the product safety approach to regulation. The stated purpose of the regulation is to "improve the functioning of the internal market" as well as "ensuring a high level of protection of health, safety, fundamental rights".[73] According to Article 26 of the AI Act, deployers (users) of high-risk AI systems must ensure that [the AI systems] are used in accordance with their intended purpose and monitor their operation.

6 THE AI ACT AND EMPLOYMENT

As the AI Act is a single, horizontal regulation it is "not specifically designed to govern the impact of artificial intelligence in the world of work".[74] Still, the AI Act is applicable to AM systems that incorporate AI. Annex III (4) enumerates specific use-cases in the employment context that are considered high-risk when AI is involved. These include: (i) AI systems intended to be used for recruitment or selection of job candidates (e.g. résumé-scoring algorithms or hiring bots), and (ii) AI systems intended to be used for promotions and terminations, task allocation, and to monitor and evaluate worker behaviour. Recital 57 argues that these systems should be classified as high-risk as they "may have an appreciable impact on future career prospects, livelihoods of those persons and workers' rights". By placing these use-cases in Annex III, the AI Act labels AM as a high-risk AI system.

An AI system is not considered high-risk if it does not pose a significant risk to health, safety, or fundamental rights, including not materially influencing

[69] Article 6(1) and Annex I AI Act.
[70] Article 6 and Annex III AI Act.
[71] M. Almada & N. Petit, *The EU AI Act: Between Product Safety and Fundamental Rights* (SSRN 2023).
[72] Article 8–22 AI Act.
[73] Recital 1 AI Act.
[74] C. Cristofolini, *Navigating the Impact of AI Systems in the Workplace: Strengths and Loopholes of the EU AI Act from a Labour Perspective*, 17(1) Italian Lab. L. EJ. 75–103, 79 (2024), doi: 10.6092/issn.1561–8048/19796.

decision-making outcomes.[75] This applies if the AI system: (a) performs a narrow procedural task; (b) improves the result of a previously completed human activity; (c) detects decision-making patterns or deviations without replacing or influencing human assessment, unless properly reviewed; (d) performs a preparatory task for assessments relevant to the use-cases listed in Annex III. However, any AI system that profiles natural persons is always considered high-risk. The definition of profiling in the AI Act follows the definition in Art. 4 (4) GDPR:

> (any form of automated processing of personal data consisting of the use of personal data to evaluate certain personal aspects relating to a natural person, in particular to analyse or predict aspects concerning that natural person's performance at work, economic situation, health, personal preferences, interests, reliability, behaviour, location or movements)[76]

Most AM systems involve processing personal data related to individuals' work performance, making them high-risk AI systems.

6.1 Risk assessment under the AI act

For AI systems designated as high-risk the AI Act imposes a comprehensive risk management obligation on providers. Article 9 of the Act stipulates providers must establish a risk management system as part of the development process. This system is to be a continuous, iterative process integrated into the design, and planned and run throughout the entire lifecycle of the high-risk AI system. The provider is required to identify and analyse the known and reasonably foreseeable risks of the system, estimate and evaluate the risks of any harm it may cause, and take appropriate and targeted risk management measures to reduce risks to an acceptable level. The AI system should also be designed in such a way "that they can be effectively overseen by natural persons during the period in which they are in use".[77]

The AI Act's risk management provisions explicitly recognize that risk assessment is not a static exercise, given the complexity of AI. For instance, Article 9(2) of the AI Act requires providers to identify the risk the system can pose when "used in accordance with its intended purpose, and under conditions of reasonably foreseeable misuse". A case such as a productivity-tracking AI that is used to exert excessive work pressure, and incentivize workers to skip breaks, thus creating a safety hazard, may now be considered reasonably foreseeable misuse.[78] The Act requires the provider to anticipate such circumstances during

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75 Article 6.3 AI Act.
76 Article 3 (52) AI Act.
77 Article 14(1) AI Act.
78 Delfanti, *supra* n. 24; Jackson, *supra* n. 26; Lorson et al., *supra* n. 24; Nilsson et al., *supra* n. 10.

development and eliminate or mitigate risks. This might involve setting default limits on work intensity metrics or ensuring the system flags possible overwork to the user, thereby helping to prevent foreseeable harms like extreme fatigue or stress-related illness.

Additionally, the risk management system must incorporate testing of the AI system throughout development to identify appropriate risk management measures.[79] Applied to AM, this could mean simulating the system's behaviour under different workplace conditions or with diverse worker inputs to see if any safety issues emerge. If risks are identified at any point, Article 9(5) requires they be eliminated or reduced to the extent possible through design changes or control measures, and any residual risks be communicated to the user.

Under this scheme, most legal obligations regarding AM system risk assessment and risk mitigation fall on the AI provider. The employer using the system is considered a "deployer" under the Act. Article 26 outlines obligations for the deployer, including using the system in accordance with the instructions, ensuring input data is appropriate, monitoring the system's operation, and keeping logs. When deployers have reason to consider that using the AI system presents a risk beyond what is reasonable and acceptable in relation to the purpose of the system, they must inform the provider and relevant market surveillance authority and stop using the system. Before using high-risk AI systems, employers must inform worker representatives and affected workers that they will be subject to the use of the system.[80]

Critically, the AI Act does not require the employer to independently assess AI's impact on worker health or rights before deployment – that prior analysis is expected to have been done by the provider. The Act states that compliance is without prejudice to other laws more favourable to workers, meaning employers remain subject to general duties under labour and privacy laws.[81] The Act stops short of imposing additional, proactive duties on employers to evaluate algorithmic risks in their specific workplace. Employers however determine targets, sanctions, staffing, pace, and monitoring intensity, and these AM deployment choices can lead to psychosocial and organizational risks. Provider-centred regulation cannot, therefore, on its own, ensure workplace risk prevention.

6.2 Documentation and instructions for use

Complementing the risk management process, the AI Act mandates extensive technical documentation and instructions to accompany high-risk AI systems.

[79] Article 9(6) AI Act.
[80] Article 26(7) AI Act.
[81] Article 2(11) AI Act.

Article 11 requires providers draw up technical documentation demonstrating the system's compliance.[82] Article 13 focuses on "instructions of use" that must be supplied to the deployer. These instructions should be "concise, complete, correct and clear".[83] The instructions must inform the user about the AI system's intended purpose, its performance (accuracy levels, limitations), and any conditions necessary for its safe and effective use.[84] Importantly, the instructions must also specify "any known or foreseeable circumstance" that could lead to risks to health, safety, or fundamental rights.[85] In practical terms, a vendor of an AM tool might include warnings such as: "This system's productivity scores should not be used as the sole basis for disciplinary action", or "Using this system to continuously track worker movements could cause stress – regular breaks and human oversight are advised". By conveying such information, the provider guides the employer to deploy the AI system in a manner consistent with safety considerations identified in development.

The AI Act's approach here mirrors established product safety regimes, where, for example, user manuals for machinery must list residual risks and proper usage to avoid accidents.[86] The AI Act effectively extends that model to AI software, proposing that if the product comes with thorough instructions and warnings, a responsible user can operate it without endangering people. However, consistent with Article 9(2), Article 13(3) (b) (iii) requires that the instructions identify not just the intended use but also the "reasonably foreseeable misuse" of the AI system, again placing the onus on the developer to predict how things could go wrong in real-world applications and to alert the user accordingly. Given knowledge on how AM is currently used, foreseeable misuse might include using the AI beyond its design scope, such as using an AI scheduling tool intended for one context in a different context without recalibration, or using it to make decisions that it is not designed to support, such as applying a task-allocation algorithm to decide layoffs. By flagging these in the documentation, the provider fulfils their duty under the AI Act, effectively passing the responsibility to the employer to heed those warnings. However, the expansive use of AI and AM across multiple work contexts and functions raises questions about the capacity for a developer to foresee every possible risk of misuse, creating a risk of responsibility-shifting: compliance may be treated as complete at the provider level while context-specific OSH risks in deployment remain insufficiently identified and managed. In practice, AI Act compliance, technical documentation, standards-based assurance, a declaration of

[82] Annex IV AI Act.
[83] Article 13(2) AI Act.
[84] Article 13(3) AI Act.
[85] Article 13(3) (b) (iii) AI Act.
[86] *See* inter alia, Art. 10.7 Regulation EU/2023/1230 on machinery.

conformity and CE marking, may be taken as proof that a system is "safe", even where workplace-specific hazards and cumulative harms have not been assessed in situ. The issue is therefore not a formal legal conflict but one of regulatory gaps and substitution in application: product-oriented compliance artefacts may substitute and thereby dilute the employer's duty to identify and manage risks that arise from management decisions about how AM is deployed.

6.3 Conformity assessment and pre-market compliance

To enforce the above obligations, the AI Act employs a compliance mechanism borrowed from EU product law: the conformity assessment. Before a high-risk AI system can be put on the market or into service, the provider must ensure it undergoes a conformity assessment procedure.[87] This culminates in the provider making an EU declaration of conformity and affixing a CE marking to the system, indicating it meets the AI Act's requirements.[88] For stand-alone high-risk AI systems, like most AM tools, the Act envisions a process of self-assessment. The provider internally verifies compliance and issues the declaration on its own responsibility.[89] As part of the conformity assessment procedure based on internal control, the provider must conduct a self-assessment to (1) verify that the quality management system (QMS) complies with Article 17 of the EU AI Act, covering areas such as regulatory compliance, technical specifications, quality controls, testing, data governance, risk management, monitoring, and reporting; (2) review the technical documentation to ensure the AI system meets the essential requirements for high-risk AI systems outlined in Chapter III, Section 2; and (3) confirm that the AI system's design, development, and post-market monitoring (as specified in Article 72) align with the contents of the technical documentation.

External third-party assessment by a notified body is reserved for certain AI systems that are already subject to existing sectoral safety regimes (e.g., AI used in medical devices or elevators, per Annex I) or for highly sensitive applications like biometric identification. This means that an AI vendor selling an AM tool would self-certify that it meets the Act's criteria. Self-certification is not unusual in EU product regulation. However, given the provider has a commercial interest in getting the product to market, without rigorous external oversight, risk assessments and mitigations might be treated as a formality. The Act does empower authorities to review documentation and enforce compliance *post-hoc*, and it also provides for post-market surveillance obligations. Article 63 requires providers to monitor the

[87] Article 43 AI Act.
[88] Article 48 and Recital 129 AI Act.
[89] Annex VI AI Act.

AI system's performance and report serious incidents or malfunction. Nonetheless, during the crucial development phase, the thoroughness of the risk assessment largely hinges on the provider's diligence and good faith, subject to potential audit.

When a "substantial modification" is made to an AI system, a new conformity assessment is required under Article 43(4) AI Act. This implies that if a deployer makes changes to an AI system which is not foreseen or planned in the initial conformity assessment by the provider, it must undergo another one, considering the changes made to the system.

Harmonized standards are expected to be of major practical importance for implementing the AI Acts requirements.[90] Compliance with such standards is not legally mandatory. However, providers whose systems conform to harmonized standards benefit from a presumption of conformity with the relevant AI Act requirements.[91] Harmonized standards are developed within the European standardization system, principally by the European standardization organizations recognized under Regulation EU/1025/2012.

The delegation of quasi-regulatory functions to standardization bodies has long been criticized, particularly in the light of EU doctrines which limit the delegation of discretionary powers and seek to prevent the transfer of essential policy choices away from institutions subject to robust legal and political accountability.[92] Critics argue that standard setting may, in practice, shape the content of legally relevant obligations while remaining subject to comparatively limited administrative and judicial controls.[93]

Regulation EU/1025/2012 provides for stakeholder participation, including representation of workers' interests through recognized organizations, but empirical work suggests that industry actors often dominate standardization processes.[94] For example, a recent analysis of CEN-CENELEC Joint Technical Committee 21 (JTC 21), reports that a majority of the members of the JCT 21 represent corporations and consultancies.[95] In the AI Act context, this raises legitimacy and governance questions about whether all affected stakeholders are adequately

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90 N. Giorgi, "Standardising AI – a trade union perspective", in A. Del Castillo Ponce (ed), Artificial intelligence, labor and society, ETUI, 2024.

91 Article 40(1) AI Act.

92 The so called Meroni-doctrine, from the Case 9/56, Meroni v. High Authority, ECLI: EU:C:1958:7. But the relevance of the Meroni doctrine has been challenged, see M. Medzmariashvili, Delegation of Rulemaking Power to European Standards Organizations: Reconsidered, 44, Legal Issues of Economic Integration, Issue 4, (2017) 353-366.

93 A. Leyden, Standards and the EU AI act: legitimacy, state of play, and future challenges. Information & Communications Technology Law, 1–31 (2025).

94 European Commission, Independent review of the European standardization system: Final report – Annexes, Publications Office of the European Union, Luxembourg, 2015.

95 Corporate Europe Observatory, 2025, Biased baked in: How Big Tech sets its own AI standards, https://corporateeurope.org/en/2025/01/bias-baked, (accessed 17 March 2026).

represented in the development of standards that may become the primary practical route to demonstrating conformity. Because standards-based conformity often becomes the dominant compliance interface, it can shape what counts as "risk" in procurement and deployment decisions, privileging measurable technical hazards over organizational and psychosocial ones.

In sum, the AI Act creates a pre-market risk governance regime for high-risk AI systems. The developer of an AM system must, *ex ante*, identify and mitigate risks, document its safety measures, and self-certify compliance as a condition for market entry. The underlying assumption is that, if this process is done correctly, the AI product will be "safe enough" for any context of use within its intended purpose (or at least, that users can be informed of any remaining risks). This approach aligns with classic product safety philosophy – the manufacturer is responsible for making a product safe by design, while the user is responsible for following instructions and using it properly. However, as the next part of this article explores, applying this model to AI in the workplace reveals significant limitations, especially when juxtaposed with the reality of how OSH risks emerge and are managed in work environments.

7 LIMITATIONS OF A PRODUCT SAFETY REGULATORY APPROACH IN ADDRESSING WORKPLACE-SPECIFIC RISKS

7.1 Product safety *v.* workplace safety: Divergent paradigms

The AI Act's regulatory logic is fundamentally that of product safety law – focusing on the properties of the technology artifact (the AI system) and its compliance with certain standards before it is sold or deployed to the European market. This paradigm works best for safety and health risks that are intrinsic to the product itself and relatively uniform across contexts: For example, a faulty machine guard or a biased facial recognition algorithm that can be identified and fixed in the design phase, or designing working time regulation into an AM scheduling system. Once fixed, the product is presumedly safe in any setting. However, many risks presented by AM systems are not solely a function of the software's design; they arise from the interaction between the technology and the specific work environment in which it is embedded.

One limitation is that a pre-market risk assessment by an AI provider can only incorporate general assumptions about the use context, whereas actual workplaces exhibit great variety. The AI Act does instruct providers to consider the "intended purpose" and "reasonably foreseeable misuse" of their systems,[96] but in practice it

[96] Article 9(2) AI Act.

is exceedingly difficult for a distant developer to predict all the relevant contextual factors that influence risk. For instance, an AI scheduling system might be designed with an average workforce in mind, but the stress impact of its schedules could depend on individual factors such as a worker's pre-existing health conditions or on local factors such as the pace of work at the facility, existing staffing levels, workforce composition and seasonal peaks. These factors differ widely between industries and workplaces, even using the same scheduling algorithm. The provider could not fully anticipate an individual employer's strategy or management style to know if, for example, the AI system will be used to issue directives or as an advisory tool, or whether aggressive or moderate performance thresholds will be set. If the product's safety depends on managerial choices, then ensuring safety cannot be accomplished by the provider alone at the design stage, even with instructions about intended use and warnings about limitations of the system to mitigate risks. To assess the significance of the risk of harm in the development phase is inherently challenging and context-dependent. Many critical questions, such as what adverse effects does AM have on the workers, how severe might violations be, what trade-offs are acceptable, cannot be answered divorced from the workplace scenario. Reviewing the possibility of AI systems achieving fairness, Veale & Binns concluded "it is impossible for a model to satisfy several of these constraints at the same time, except in exceptional cases which are unlikely to hold in the real world".[97]

Another limitation of the product safety approach is the difficulty capturing gradual or cumulative harms in a pre-market assessment. Traditional product safety often focuses on acute failures (e.g. a machine malfunction causing injury) or obvious hazards (e.g., the toxicity of a chemical). AM, however, can give rise to long-term health effects from psychosocial risks. As noted earlier, AM's constant surveillance or excessive workload demands can contribute to increased stress, anxiety, fatigue, or mental health deterioration.[98] These harms do not constitute a singular "incident" with acute health effects that would be identified during testing; instead, they "can accumulate without a single event tripping a threshold of seriousness".[99] An AI system might pass all functional tests and never produce an error, yet still pressure employees to work faster, slowly eroding their health. In the AI Act context, a provider might struggle to decide if such a slow-burn effect constitutes a "significant risk" triggering design alterations – especially when harm severity can vary individually and be moderated by other individual, social or organizational factors. Unless a risk is obvious and quantifiable in advance, it may

[97] M. Veale & R. Binns, *Fairer Machine Learning in the Real World: Mitigating Discrimination Without Collecting Sensitive Data*, 4(2) Big Data & Soc'y 3 (2017) doi: 10.1177/2053951717743530.
[98] Delfanti, 2021; Doellgast et al., 2023; Lorson et al., 2023; Nilsson et al., 2025a.
[99] Cristofolini, *supra* n. 66, at 85.

be underestimated or ignored in the conformity assessment. This could be exacerbated if the AI Act's definition of high-risk ("significant harmful impact") is interpreted too narrowly and is taken to mean only immediate physical injury.[100] Without clear guidance, some AI providers might considert psychosocial hazards outside the scope of "safety" considerations.

Moreover, the self-assessment nature of the conformity process raises concerns about thoroughness and potential conflicts of interest. Self-assessment is not inherently flawed; many responsible companies will do due diligence. However, the Act fails to provide sufficient guidance on how the self-assessments should be made,[101] and with "such vague stipulations [about data sets being "relevant", "representative" and "free from errors"] it is possible self-assessments could become rubber stamps".[102] In the absence of an external audit, providers may be incentivized to interpret requirements in the most convenient way – presuming that no "foreseeable misuse" exists beyond what it has already handled, or that any workplace harm is due to (unforeseeable) misuse by the deployer and thus outside their responsibility. Compliance may be treated as a tick-box exercise, documenting some generic risks and remedies that look acceptable on paper, without truly engaging with the messy realities of workplaces or making the product safe for all deployment contexts. This is sometimes described as the problem of "marking one's own homework"[103] in regulatory compliance – the process lacks an independent check.

Finally, product safety regulation typically assumes a clear delineation between the manufacturer's responsibility (produce a safe product) and the user's responsibility (use it correctly). Yet in AM, this line is blurred: the service provided by the AI is essentially managerial action itself. The employer is not just using a tool but delegating managerial decision-making to that tool. Many of the risks outlined earlier stem from how that delegation is structured and governed. In other words, the risk is co-produced by the technology and the employer's management strategy. If an employer chooses to override safety warnings in the AI's instructions or pushes the AI to maximise productivity at the expense of worker welfare, the harm results from a combination of the AI's capabilities and the employer's choices. Similarly, harm may arise if the AI is tailored to the organization and built/trained on employer provided data based on pre-existing harmful management practices (such as excessive workloads) unknown to the

[100] Although, as mentioned above, even systems that do not risk 'significant harmful impact' are considered high risk systems if the system involves profiling of natural persons.

[101] N. A. Smuha, E. Ahmed-Rengers, A. Harkens, W. Li, J. MacLaren, R. Piselli & K. Yeung, *How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission's Proposal for an Artificial Intelligence Act* (2021), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3899991 (accessed 8 Jan. 2026).

[102] K.-L. Aislinn, *Dispatch No. 39 – European Union – 'The AI Act and Algorithmic Management'*. Comp. Lab. L. & Pol'y J. 7 (2021), https://cllpj.law.illinois.edu/dispatches (acccessed 8 Jan. 2026).

[103] *Ibid.*, at 6.

designer. A product-centric regime might classify the product as technically 'safe' if a warning has been given by the developer and deem the employer's misuse or biased training data as the problem.

From a worker's perspective, this distinction offers little comfort or protection from workplace hazards arising from AM. Regulation should prevent or deter misuse regardless of where it originates or ensure accountability for it. The AI Act's approach to misuse is front-loaded on the provider to predict and warn, leaving general laws (labour law or tort law) to catch any misuse issues. This gap is problematic because it places heavy reliance on non-AI-specific regimes to handle what may be novel or opaque risks. Labour laws impose duties on employers that could encompass such misuse, but the integration between those duties and the AI Act's regime is not clearly delineated, potentially leading to enforcement difficulties.

7.2 Contextual and dynamic nature of OSH risks in Algorithmic Management

Many AM risks are emergent and dynamic. Risks may not be fully apparent until the system is deployed and interacts with human behaviour. Workers might initially adapt to a new AI tool in unexpected ways – for example, by speeding up work unsustainably to meet AI-generated goals or finding shortcuts that create new safety hazards. The risk management performed during development might not anticipate these adaptations even if systems are tested to identify "the most appropriate and targeted risk management measures".[104] In safety-critical industries, there is recognition that real-world use can reveal unanticipated risks, making feedback and post-market surveillance important. The AI Act does include post-market monitoring by providers,[105] but as a form of product surveillance where the provider tracks performance issues or malfunctions. It does not formalize mechanisms whereby workers' experience of being subjected to the AI system would feed into improving the AI's risk management. Put differently, while a medical device manufacturer must track adverse events in the field and update its devices accordingly (*see* inter alia, Regulation EU/2017/745 on medical devices), an AM vendor might be unaware its system is indirectly contributing to employee burnout at a company – unless the employer reports it. The iterative improvement of AI safety might stall if there is no structured way to learn from contextual deployments. This underscores that pre-market assessment, done once, is

[104] Article 9(6) AI Act.
[105] Article 72 AI Act.

insufficient for dynamic systems whose risks evolve with usage. Instead, continuous, context-specific assessment is needed.

Additionally, the AI Act's concept of "intended purpose" could be too narrow when it comes to workplace use. Article 3(12) defines intended purpose as the use for which the system is meant according to the provider. However, in employment, even a system not specifically intended to manage workers can end up being used in ways that have consequences for workers. An AI technology developed to improve driver safety by monitoring driver fatigue could be repurposed by an employer to monitor worker performance or behaviour.[106] Providers might avoid acknowledging an employment use-case to escape high-risk classification – a loophole wherein they "claim their applications could pose risks to workers but classify them as non-significant" by pointing to a benign intended purpose.[107] This is a regulatory blind spot: the Act does not address who bears responsibility when AI is used in a high-risk way that the provider did not intend. The AI Act states that if a deployes "modify the intended purpose of an AI system … in such a way that the AI system concerned becomes a high-risk AI system" they will be considered a provider of a high-risk AI system and subject to the responsibilities of providers.[108] Yet, when use outside the intended purpose is considered a modification of the intended purpose is unclear. The likely outcome is inconsistent handling of such scenarios, potentially leading to risky use of AI systems.

Finally, the product safety approach struggles with human agency and organizational measures – both central concepts for mitigating workplace risks. Under OSH-regulation, if a tool is potentially hazardous, the employer can mitigate risk by choosing a safer tool or implementing procedures, training, or limits on its use. If a machine can cause strain, the employer might impose work-rest cycles or job rotation. Thus, if AM intensifies work, an employer could counterbalance it by capping the number of tasks assigned per hour or ensuring additional break time. These workplace policies are not things an AI vendor can enforce. The AI Act's model assumes the AM product will build in all necessary safeguards. Realistically, some AM safeguards are best implemented as organizational controls. A product-centric regime might declare the product "safe" if used properly, but without requiring the deployer to adopt complementary measures, safety is theoretical.

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[106] M. Kullman & A. Cefaliello, *The Interconnection Between the AI Act and the EU's Occupational Safety and Health Legal Framework* (G. EU Law, Labour law, Privacy 2021), https://global-workplace-law-and-policy.kluwerlawonline.com/2022/01/24 (accessed 8 Jan. 2026).

[107] Cristofolini, *supra* n. 74, at 85.

[108] Article 25(1) AI Act.

7.3 Foreseeable misuse and the limits of *ex ante* control

The AI Act's obligation on providers to anticipate "reasonably foreseeable misuse" is similarly limited by employer behaviour. Many negative outcomes may arise not from mishandling of the AI tool but through misuses of managerial power via the tool. For example, an algorithmic tracker might be functioning as designed but is used to harass or set unrealistic performance demands. Even with a warning provided in the tool's instructions, preventing such misuse requires enforcement of labour standards (ensuring employers do not abuse the tool) or technical locks (designing the AI such that it cannot be used to, say, continuously monitor without breaks). The latter might be possible in some cases (e.g., AI could automatically suspend monitoring after certain hours) but might not be welcomed or circumvented by buyers. Thus, the product safety approach is limited to making the AI capable of safe use and cautioning against unsafe use. It cannot ensure ethical, context-aware deployment.

It is also possible some AI-related OSH risks might not be viewed as "safety" issues by a tech provider. For example, as seen with precarious work, an algorithm that ranks workers could create psychosocial harm by fostering competition or increasing fears of job losses.[109] Tech vendors might consider this a management not a safety issue. Even with the AI Act's general language nudging them to think of "health and safety risks" and fundamental rights, tech providers may lack the expertise to evaluate psychosocial workplace hazards. They are unlikely to have ergonomists or occupational psychologists on their product development teams. This expertise gap means many providers will not deeply engage with OSH risk mitigation unless guided or compelled by specific regulations or input from workers or their representatives.

In summary, merely applying a product safety-style framework to workplace AI reveals inherent limitations. It presumes a level of predictability and control over context that is at odds with the variable and evolving nature of workplaces. It places heavy responsibility on designers to get things right in advance, even for outcomes that only manifest in use, and it does not fully resolve how to handle misuse or abuse by the employer or employees. These limitations suggest that while the AI Act provides an important baseline, it may leave significant gaps in protecting workers – gaps that traditional OSH regulation is better suited to fill. The next section will elaborate on applying OSH regulation to mitigate risks of AM use, highlighting the need for a complementary relationship between the AI Act and labour law mechanisms.

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[109] Bluff et al., *supra* n. 20; Nilsson et al., *supra* n. 10.

8 THE LABOUR LAW PERSPECTIVE: OSH FRAMEWORKS *VS.* THE AI ACT'S APPROACH

OSH legislation, both at the EU level and within Member States, imposes a duty of care on employers to ensure the safety and health of workers "in every aspect related to the work".[110] This fundamental principle is broad by design: it covers physical safety, mental health, and any other aspect of working conditions that could affect wellbeing. OSH law embraces a preventive and participatory approach. Employers are required to conduct regular risk assessments of their workplaces and to put in place measures to eliminate or reduce identified risks, following a hierarchy of prevention – avoid risks, evaluate those that cannot be avoided, combat risks at source, adapt work to the individual, and more.[111] When new technology or work processes are introduced, the employer must evaluate any associated risks. Workers or their representatives have a right to be informed and consulted about OSH risks and measures and to appeal to authorities if they believe the prevention measures are inadequate.[112] In sum, OSH regulation is contextual-because it looks at the actual workplace and dynamic because it requires ongoing assessment and adaptation as conditions change.

Applying OSH principles to AM would require an employer to proactively assess the ways in which any AM system deployed to manage workers will impact the work environment and worker health and safety. While the AI Act does not mandate it, however the employer's general duty is likely akin to conducting an "algorithmic risk assessment" before and during the use of the tool. The Framework Directive is flexible enough to encompass digital risks. For instance, Article 6(2) (d) of the Directive instructs employers to adapt work to the individual, particularly regarding the design of workplaces, the choice of work equipment, and the pace of work, "with a view, in particular, to alleviating monotonous work and work at a predetermined work-rate". This is directly relevant to AM because if an AM system imposes a predetermined pace, the employer has a duty to ensure this does not harm employees, including modifying work processes – perhaps by adjusting targets or allowing pauses – to alleviate that risk. Similarly, if the AM system increases monotony or reduces workers' control, the employer should consider OSH interventions such as job rotation or task variation to mitigate potential negative effects.

Unlike the AI Act, OSH regulation puts the burden on the party with direct control over the work setting – the employer.[113] This makes practical sense

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110 Article 5 Framework Directive. Article 5 Directive 89/391/EEC.
111 Articles 6 and 9 Directive 89/391/EEC.
112 Articles 10 and 11 Directive 89/391/EEC.
113 A. Paliokaitė, A. Christenko & A. Aloisi et al., *Study Exploring the Context, Challenges, Opportunities, and Trends in Algorithmic Management in the Workplace – Final Report* (European Commission: Directorate-

because the employer can implement tailored measures, such as introducing AM rules and management practices to limit work hours, mandate breaks or negate the intrusive use of AM. Failing to take such steps might violate the employer's OSH duties if the use foreseeably impairs worker health. As existing OSH duties already apply to psychosocial impacts of algorithmic monitoring, independent of AI-specific laws, employers are responsible for preventing AM risks under current OSH law.[114]

9 CONCLUSION

As the use of AM has expanded well beyond the platform economy, and AI has increasingly been integrated into AM technologies, attention has been drawn to the OSH risks that arise. While AM has potentially positive OSH applications, in most instances the implementation of AM has been shown to have negative impacts. AM can lead to work intensification, contributing to increased stress, new physical hazards, and encourage risk-taking at work. Many potential OSH risks arising from AM are, however, difficult to foresee because they may be context-specific and exacerbated or mitigated by management strategy, practices and implementation processes. Risk assessment is a central requirement in existing OSH regulation, and features in the new AI Act, created to regulate the development and deployment of AI technologies within the EU. This study critically examined the risk assessment requirements of the AI Act, discussing its suitability for managing OSH risks, and highlighting the gaps that arise from the proposed product safety approach, especially when compared to existing OSH frameworks.

The analysis identifies that risks posed by AM systems are highly context-dependent and many of the OSH adverse effects on workers, cannot be adequately identified or mitigated through pre-market assessments alone. Psychosocial risks are often cumulative, gradual, and shaped by organizational practices, making them difficult to capture in standard product-based compliance mechanisms. Furthermore, the AI Act's assumption that providers can anticipate all reasonably foreseeable misuse does not reflect the dynamic and diverse nature of workplaces.

The reliance on self-assessment and harmonized standards, combined with limited employer obligations, raises concerns about the adequacy of the AI Act to protect workers. Unless the employer is proactive in integrating AM into their

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General for Employment, Social Affairs and Inclusion, Visionary Analytics, Publications Office of the European Union 2025), https://data.europa.eu/doi/10.2767/5629841 (accessed 8 Jan. 2026).

[114] Thorn Jensen et al., *supra* n. 15, *see also* inter alia, Arbetsmiljöverkets föreskrifter om organisatorisk och social arbetsmiljö, 2 kapitlet AFS 2023:2 (Swedish Work Environment Agency Provisions on Organisational and Social Work Environment, 2 Chapter AFS 2023:2.

OSH procedures, key risks may go unaddressed. This creates a regulatory gap: while AM systems are classified as high-risk under the AI Act, the regulation assumes that proper use will guarantee safety, without requiring the employer to validate this through workplace-specific evaluation. That many AM-related risks cannot be fully predicted or mitigated in the design stage underscores the need for complementary workplace-specific risk assessments.

For AM technology developers who wish to integrate AI into their systems, this regulatory gap has significant implications, as it places the burden of OSH management on the developers, who are required to design the system and provide instructions for use that identify and mitigate any "foreseeable risks", including "foreseeable misuse". In practice, however, AM is a socio-technical system and there are many context-specific factors that may influence the ways in which an AM system is implemented and used that have ensuing OSH implications. Management strategies, workforce composition, union presence, the pace and seasonality of work, and the intensity of inter-firm competition all shape how AM impacts worker health and safety. These factors are outside the control of the developer. A product safety approach to regulation does not account for these factors. In contrast, OSH frameworks focus on the context within which AM will occur by placing the onus on employers to use the tools in ways that will minimize harm to workers.

To address these challenges, this article argues for a closer alignment between the AI Act and OSH regulation. This includes recognizing that algorithmic systems deployed in employment contexts are not only technical artefacts but also instruments of managerial control, with significant implications for working conditions. It also requires that employers interpret their OSH duties to include the impact of AM systems and that regulatory authorities provide clearer guidance on how to conduct such assessments. It is important to acknowledge that an AM system is not only a product (to be regulated as such), but a technology with effects highly dependent on the social and organizational context in which it is adopted. Technologies are never neutral, and the design and the use of them are rooted in the dominant social values.[115]

By its very nature the AI Act establishes a generic approach to deal with an array of harm that may arise and assumes these can be addressed purely in terms of design parameters. As emphasized in this paper this approach is too generic and fails to address the critical process dimensions of OSH or indicate how controls could be made effective across an array of industries with different hazard profiles. At the same time, OSH regulation needs to be reconfigured to better address AM-related

[115] I. Ajunwa, *The Quantified Worker: Law and Technology in the Modern Workplace* (Cambridge: Cambridge University Press 2023).

challenges including mandating regulator and union access to algorithms and encouraging negotiated industry level codes of practice or AM guidelines.

In conclusion, the safe deployment of AM systems in the workplace cannot be achieved through product regulation alone. The AI Act sets important *ex ante* safeguards but must be complemented by robust, context-sensitive risk management practices anchored in existing OSH frameworks. Future policy development should support stronger integration between AI governance and labour protection regimes, promote institutional oversight, and ensure meaningful worker participation in the deployment and monitoring of AI-based systems. Only then can AM be deployed in a way that safeguards, rather than undermining, worker health, safety, wellbeing, and dignity.

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